

Depending on how it is processed, silk can be made to breakdown inside the body almost instantly or to persist for years. This technique offers incredible opportunities for medical applications, but we should not underestimate the power of using such devices to improve our understanding of human performance.

In the near future, we would be able to measure in real-time how our body responds to a variety of exercise paradigms as well as nutritional interventions, improving our understanding of human biology and giving us a better chance to prescribe exercise programmes.

Sensitronics

Engineers are not only developing athletic technology that can be worn on the body but also technology that is incorporated into sports equipment itself. One example is a hi-tech smart soccer ball that contains a number of internal sensors. When the ball is kicked, these sensors gather data on the speed and spin of the ball as well as its trajectory and strike point. The accompanying app includes tutorials and challenges that can be used to record and improve progress during training.

Engineers have also designed a smartracquet equipped with sensors that can give players information about the number of strokes, power, impact and technique. These technologies employ Bluetooth to transmit information to a corresponding smartphone app for tracking and analysis.

Sensitronics can play a great role in the design and working of sports equipment. Force sensing resistor (FSR) technology can be used to enhance any game or outdoor activity.

For example, FSRs can accurately detect position, monitor impacts, measure gripping force and provide speed control for trolling motors. These are made of a polymer thick film ink, typically screen printed on Mylar (PET) film, depending on the requirements of the application. As



Baseball and new communication technology

force is applied to the device, electrical resistance decreases.

The ink formulation can be customised for application-specific requirements such as minimising saturation with greater force as well as for very low force needs. Temperature, humidity and shear are some of the considerations. The FSR can be used for applications such as computer input devices, musical instruments, interactive toys, robotics, fingertips, automotive, sports or medical applications.

Seemingly low-tech gadgets like golf clubs are benefiting from electronics technology advancements. These are now being designed to save energy. There are thousands of special energy capsules that have been melted together into one midsole.

High-speed cameras are being utilised more frequently in the wide world of sports. One application is to help referees and sports officials make less subjective calls about plays on the field.

Camera based goal line technology is commonly used at present in many games like football, hockey, lawn tennis and badminton. Cameras are placed high-up at different points around the sports venue. If a ball crosses the goal line during a match, the word GOAL is transmitted to a smartwatch worn by the referee. The system is able to determine whether

a goal was viable within one second, and to an accuracy of 0.5cm.

SportVu is a statistical tracking technology currently used in basketball. It also uses high-speed cameras. The cameras, which are affixed to the rafters of the arena, capture movements of players and the ball 25 times per second in X, Y and Z coordinates. This information is then transmitted to software that calculates the players' dribbles, passes and touches.

Coaches can use this information to make decisions on the court and assess whether players are in good physical condition to perform. Data could also be used to identify players' strengths and weaknesses and for scouting new recruits. SportsVu also enables players' stats to be shared with both viewers and fans.

Conclusion

It is challenging to package electronics with sports in an aesthetically pleasing but ergonomically small and lightweight enclosure. Applications of electronics in sports include getting in shape (fitness), playing sports professionally (accuracy and performance) or watching sports (recording and transmission).

Technology with electronics in sports is offering new insights and interactivity to the world of sports. These powerful tools are putting eye-opening real-time knowledge into the hands of users like never before. Even mobile apps have the ability to maximise an athlete's potential, help teams reach new heights or players experience sports in fun new ways.

Athletes and gamers of tomorrow will need engineers and computing professionals to develop new applications that marry a passion for sports and cutting-edge electronics based technologies. As sports technologies evolve, these will continue to help sports persons everywhere push the boundaries of physical excellence. ●